

SECTION 01 51 24
TEMPORARY STEAM GENERATING FACILITIES AND CONTROLS

PART 1 GENERAL

1.1 DESCRIPTION

- A. The requirements of this Section apply to all Division 01 and Division 21, 22, and 23 sections. This section generally governs work related to the temporary boiler plant trailers, temporary control rooms, and related piping, electrical, controls, and other associated facilities.
- B. Provision of temporary steam generating facilities and controls includes, but is not limited to, the following elements. The contractor remains responsible for a fully operational temporary system that can provide steam to the hospital in accordance with the performance conditions listed in this Section.
 - 1. Mobile, trailer enclosed, steam boiler plants.
 - 2. Control room with restroom and break room provisions.
 - 3. Restroom sanitary service.
 - 4. Access platform between control room and mobile, trailer enclosed, steam boiler plants.
 - 5. Piping supports, racks, and thermal stress management devices and design.
 - 6. Piping, fittings, valves, insulation, exterior jacketing, and labeling of temporary piping.
 - 7. Steam traps and condensate control systems.
 - 8. Domestic water supply, backflow preventors, piping, fittings, and valves.
 - 9. Exterior painting.
 - 10. Electrical service to all equipment requiring power.
 - 11. Coordination of delivery and receiving, rigging, installation, adjacent temporary facilities not in contract, fuel delivery of natural gas and fuel oil, operator relocation from existing control room to temporary control room, service and maintenance of temporary equipment, and all other construction duties required to ensure a fully operational temporary system.
- C. Acceptance - The following must have occurred and be documented in a letter from the Contractor to the COR prior to final acceptance and turnover for operations:
 - 1. The Contractor performs a final inspection to verify that all equipment has been installed in accordance with their requirements. The supplier shall be responsible for installation as defined herein.
 - 2. The Temporary Steam Generating Facility has been successfully started-up and has met the performance requirements included in this specification. All documentation associated with performance verification has been submitted.
 - 3. The Contractor has provided and completed training in accordance with the requirements included in this specification.
 - 4. The Contractor has turned over all Operating and Maintenance Manuals to the COR and VA Facility Operations Staff.
 - 5. The Contractor has provided acceptance testing of electrical equipment provided with the Temporary Steam Generating Facility and has included test reports with the Operation and Maintenance Manuals.
 - 6. The COR has provided written VA acceptance of submitted documentation and explicitly indicates that the VA is ready and prepared to assume operation of the Temporary Steam Generating Facility in place of the existing systems.

1.2 RELATED WORK

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.
- B. Section 01 00 00, GENERAL REQUIREMENTS.
- C. Section 01 31 00, PROJECT MANAGEMENT AND COORDINATION
- D. Section 01 32 16.15, PROJECT SCHEDULES

- E. Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.
- F. Section 01 35 26, SAFETY REQUIREMENTS
- G. Section 01 42 19, REFERENCE STANDARDS
- H. Section 01 45 00, QUALITY CONTROL
- I. Section 01 74 19, CONSTRUCTION WASTE MANAGEMENT
- J. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.
- K. Section 01 91 00, GENERAL COMMISSIONING REQUIREMENTS.
- L. Section 02 82 11, TRADITIONAL ASBESTOS ABATEMENT.
- M. Section 03 30 00, CAST-IN-PLACE CONCRETE.

1.3 APPLICABLE REFERENCES

- A. The publications are referred to within the text by the basic designation only.
- B. AMERICAN WATER WORKS ASSOCIATION (AWWA)
 - 1. AWWA C511 (2017; R 2021) Reduced-Pressure Principle Backflow Prevention Assembly
- C. NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
 - 1. NFPA 70 (2023; ERTA 4 2023) National Electrical Code
 - 2. NFPA 241 (2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations
- D. AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
 - 1. BOILER AND PRESSURE VESSEL CODE (BPVC)
 - a. Section I – Power Boilers

1.4 SUBMITTALS

- A. Submit in accordance with specification Section 01 33 23 - SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.
- B. Plans shall be developed in coordination with and compliance with Section 01 00 00 GENERAL CONDITIONS.
- C. Site Plan: Show temporary facilities, utility hookups, staging areas, and parking areas for construction personnel.
- D. Erosion- and Sedimentation-Control Plan: Show compliance with requirements of EPA Construction General Permit or authorities having jurisdiction, whichever is more stringent.
- E. Moisture-Protection Plan: Describe procedures and controls for protecting materials and construction from water absorption and damage.
 - 1. Describe delivery, handling, and storage provisions for materials subject to water absorption or water damage.
 - 2. Indicate procedures for discarding water-damaged materials, protocols for mitigating water intrusion into completed Work, and replacing water-damaged Work.
 - 3. Indicate sequencing of work that requires water, such as sprayed fire-resistive materials, plastering, and terrazzo grinding, and describe plans for dealing with water from these operations. Show procedures for verifying that wet construction has dried sufficiently to permit installation of finish materials.
- F. Dust- and HVAC-Control Plan: Submit coordination drawing and narrative that indicates the dust- and HVAC-control measures proposed for use, proposed locations, and proposed time frame for their operation. Identify further options if proposed measures are later determined to be inadequate. Include the following:
 - 1. Locations of dust-control partitions at each phase of work.
 - 2. HVAC system isolation schematic drawing.
 - 3. Location of proposed air-filtration system discharge.
 - 4. Other dust-control measures.

- G. Pipe Rack Systems: Submit coordination drawings and product data showing system layout, calculated loads and design conditions, foundations, coatings, and anticipated piping, conduit, and other mounted systems arrangement.
- H. Pipe Stress Analysis Report. Include hanger designs, anchor designs, loads, forces, and moments.
- I. The following shall be provided for all temporary equipment to be installed as part of the Temporary Steam Generating Facility.
 - 1. Complete bill of materials. The list shall include the component type (steam boiler, steam boiler, boiler auxiliary equipment pump, motor starter, motor control center, variable speed drive, panelboard, transformer, luminaire, valve, etc), model number and manufacturer.
 - 2. General Arrangement Drawing (Plans and Elevations) showing the location of all components. GA drawings must show the location, size and type of all utility connection points (Steam, condensate, natural gas, fuel oil, condenser water, domestic water, electric, controls, telecommunications, fire alarm, security, etc).
 - a. Drawings shall be to scale on E Size (30" x 42") paper. Scale shall not be less than $1/8" = 1'-0"$.
 - 3. Process and Instrumentation Diagrams for Steam System, Condensate System, Steam Boiler System, Auxiliary Steam Systems, Fuel Systems, and other major systems.
 - 4. Electrical Single Line Diagram(s).
 - 5. Electrical Lighting, Power, and Signal General Arrangement drawings showing the locations of electrical equipment, devices, and luminaires.
 - 6. Starter types, complete with shop drawings and detailed product data.
 - 7. Variable Speed Drives (if applicable) including complete product data, show drawings, and wiring diagrams.
 - 8. Loading Diagram - Loading diagram shall show proposed loading of equipment modules. This must include forces and moments at all loading points.
 - 9. Estimated Sound Power Ratings - sound power ratings (dBA) at 25 feet in all directions.
- J. Trailer Mounted Boiler Room
 - 1. Plant Design Information
 - 2. Boilers
 - a. Boiler Manufacturer
 - b. Boiler type - principle of operation
 - c. Predicted Boiler Performance Data: at site altitude, at recommended percent excess air (not to exceed 15%), for output levels of 12.5, 25, 50, 75, and 100 percent of maximum required boiler capacity. Data must include fuel and steam flow, flue gas temperature at boiler outlet, efficiency, windbox and furnace pressures, combustion airflow, excess air, predicted radiation losses, unaccounted losses, heat input rate of effective radiant heating surface of the furnace, volume heat input rate of effective radiant heating surface, NO_x and CO emissions, and flue gas oxygen level. All predicted performance data shall be based on the boiler and burner provided as an entire package.
 - d. Predicted Boiler Design Data:
 - 1) Maximum Allowable Working Pressure
 - 2) Maximum Allowable Temperature Rating
 - 3) Design Working Pressure
 - 4) Design Working Temperature
 - 5) Predicted Turndown Ratio
 - 6) Amount of heating surface, combustion volume.
 - e. Burner flow and fuel pressure data:
 - 1) Main burner fuel pressures and flows at maximum required firing rate.
 - 2) Ignitor (pilot) fuel flow and burner pressure.
 - 3) Natural gas main fuel pressure at outlet of burner-mounted pressure regulator.
 - 4) Ignitor (pilot) fuel pressure at outlet of burner-mounted pressure regulator.

- (a) Boiler Stack Data including diameter, height above building, and height above grade.
- 3. Packaged Feedwater Unit
 - a. Proposed Unit Manufacturer
 - b. Equipment Capacity
 - c. Piping Connections
 - d. Electric Power Requirements
 - e. Controls
- 4. Packaged Condensate System
 - a. Proposed Unit Manufacturer
 - b. Equipment Capacity
 - c. Piping Connections
 - d. Electric Power Requirements
 - e. Controls
- 5. Plant specialties.
- K. Valves
 - 1. Provide a valve schedule for all manual and control valves in each system. A separate valve schedule shall be prepared and submitted for each system. Schedule shall indicate, at a minimum, the following:
 - a. Manufacturer
 - b. Size
 - c. Materials of Construction
 - d. Design pressure and temperature
 - 2. Single-Source Responsibility: All valves of same type shall be from one manufacturer.

1.5 QUALITY ASSURANCE

- A. Electric Service: Comply with NECA, NEMA, and UL standards and regulations for temporary electric service. Install service to comply with NFPA 70.
- B. Tests and Inspections: Arrange for authorities having jurisdiction to test and inspect each temporary utility before use. Obtain required certifications and permits.
- C. Accessible Temporary Egress: Comply with applicable provisions in [the U.S. Architectural & Transportation Barriers Compliance Board's ADA-ABA Accessibility Guidelines] [and] [ICC/ANSI A117.1].

1.6 PROJECT CONDITIONS

- A. General and Sequencing
 - 1. The Bruce W. Carter Department of Veterans Affairs Medical Center is currently served by two, skid mounted, rental boilers located in the construction area of the Building 44 expansion. This equipment must be removed prior to beginning work on the building expansion in this contract. The rental contract and responsibility for operations and maintenance of this equipment is held by the VA. The VA will retain control of this until such time that the Temporary Steam Generating Facility specified herein, is operational and accepted by the VA. At such time, the Contractor shall assume responsibility for final shutdown and demobilization of the existing temporary equipment in coordination with the rental equipment supplier. This is intended to provide the contractor with control of steam phasing and transfer from one facility to the other and its position and impact on the overall construction schedule.
 - 2. The rental period for the Temporary Steam Generating Facilities shall be no less than the required period to perform all construction activities up to and including commissioning of the new steam plant (building 44 expansion). Removal of the equipment must be coordinated with the COR and facility operations staff.
 - 3. Include, in the overall project schedule, transition of staff to the new control room, transfer of steam generation responsibility from the Temporary Steam Generating Facilities to the

new boiler plant, demobilization, site restoration, and removal of any and all non-rental/semi-permanent facilities and utilities back to their original point of connection.

B. Existing Systems

1. The existing Building 44 steam plant boilers are not functional. The following equipment is operational and supports the existing rental boiler operation:
 - a. Deaerator and Feedwater Pump Package
 - b. Fuel Oil Day Tank and Pumps
 - c. Temporary Control Room (located on first floor, below steam boiler room)
 - d. Condensate Surge tank (located in Building 1 mechanical room)
 - e. Control Panel, located in boiler room.

C. Performance Requirements

1. N+1 Steam Capacity (PPH): ~~23,000~~13,000, Net to Hospital at Building 1 flange (Anchor A-1-1)
2. Steam Pressure (PSIG): 100 PSIG, at Maximum Continuous Rate (MCR)
3. Design Condensate Return (GPM): 0 (Zero)
4. Normal Condensate Return (GPM): 50 Percent or Less
5. Average Condensate Return Temperature (Deg F): ~~450-480~~170
6. Natural Gas Supply Pressure (PSIG): 5, at the discharge of the regulating station
7. Fuel Oil System Pressure (PSIG): 30, at Building wall. Deadhead is approximately 42 PSIG.
8. Fuel Oil System Flowrate (GPM): 0-40 at deadhead pressure.
9. Domestic Water Pressure (PSIG): Approximately 30, at the backflow preventor.
- 9-10. Average Domestic Water Temperature (Deg F): 70 or warmer depending on location and sun expose on the piping rack.

PART 2 PRODUCTS

2.1 TEMPORARY FACILITIES

- A. Field Office, General: Prefabricated or mobile units with serviceable finishes, temperature controls, and foundations adequate for normal loading.
- B. Control Room-Use Field Office: Of sufficient size to accommodate needs of the VA and steam plant operation activities. Furnish and equip offices as follows:
 1. Furniture required for Temporary Steam Generating Facility documents including file cabinets, plan tables, plan racks, and bookcases.
 2. Operating Computer Countertop directly behind window looking out at temporary boiler trailers. Desk shall be capable of holding two computers and four 27 inch computer screens.
 3. Provide electrical power service and 120-V ac duplex receptacles, with no fewer than one receptacle on each wall. Furnish room with conference table, chairs, and 4-foot- (1.2-m-) square tack and marker boards.
 4. Drinking water and private toilet.
 5. Breakroom including Dish Washing Sink, full size Refrigerator, 1100 watt minimum Microwave, 12-cup Coffee machine, and supplies.
 6. Heating and cooling equipment necessary to maintain a uniform indoor temperature of 68 to 72 deg F (20 to 22 deg C).
 7. Lighting fixtures capable of maintaining average illumination of 20 fc (215 lx) at desk height.
- C. Mobile, Trailer-mounted, Boiler Room: Of sufficient size to provide the required steam generating capacity listed in Part 1 of this Section. The boiler room shall contain the following elements and equipment:
 1. Factory-assembled and packaged, Dual-Fuel, Watertube or Firetube steam boiler. Boiler shall be designed to 200 PSIG MAWP. Fuel shall be Natural Gas and No. 2 Fuel

Oil. Assembly shall be constructed and stamped in accordance with Section 1 of the ASME Boiler and Pressure Vessel Code and be registered with the National Board.

- a. Regardless of fuel input rating, the equipment, installation, and operation shall conform to NFPA 85 for a single burner arrangement.
- b. Turndown: Boiler shall be capable of 5:1 turndown or greater
2. Condensate Receiver and Condensate Pumps
3. Feedwater Deaerator and Feedwater Pumps
- 3-4. Steam pressure reducing station for deaerator
- 4-5. Water Softeners and associated Boiler Water Chemical Treatment System. Coordinate with the COR for required chemicals and feed locations. They will be the same or similar to the new plan design including number and type of chemicals.
- 5-6. Blowdown tank with aftercooler/quench system. Aftercooler: Provide and install aftercooler assembly for automatic control of drain water temperature (Water at or above 140°F shall not be discharged from the blowdown tank to the sanitary system). Assembly shall include non-potable cold water inlet, electronic temperature regulating valve and associated equipment drain temperature sensing device, strainer, and bimetal thermometer with 2" dial.
- 6-7. Include an NFPA compliant main gas shut off valve with pull stations inside of trailers to permit emergency shutdown of the boiler plant system by an operator.

2.2 EQUIPMENT

- A. Fire Extinguishers: Portable, UL rated; with class and extinguishing agent as required by locations and classes of fire exposures.
- B. Piping Rack System: Basis of Design is ErectaRack Pre-Engineered pipe rack systems in a 3-Tier configuration.
 1. Racks shall be hot dipped galvanized steel.
 2. Provide arrangement shown in the Mechanical Drawings and modify as necessary to comply with this Section and the final equipment locations, sizes, and type.
 3. Design racks to support piping, conduit, lighting, and other mounted elements. Structural design conditions shall be per Structural Drawings and indicated seismic and wind load conditions.

2.3 PIPING, INSULATION, VALVES, FITTINGS, AND SPECIALTIES

- A. Provide piping, insulation, valves, fittings, and specialties in accordance with other Division 23 sections specifying these items.
 1. Section 23 07 11, HVAC AND BOILER PLANT INSULATION for piping and equipment insulation.
 2. Section 23 10 00, FACILITY FUEL SYSTEMS for fuel oil piping and specialties.
 3. Section 23 11 23, FACILITY NATURAL-GAS PIPING for natural gas piping and specialties.
 4. Section 23 21 11, BOILER PLANT PIPING SYSTEMS for steam, condensate, blowdown, blow-off, and associated system piping and specialties.
 5. Section 23 21 13, HYDRONIC PIPING for other non-steam systems.
 6. Section 23 22 13, STEAM AND CONDENSATE HEATING PIPING for specific steam and condensate heating piping.
- B. Submittals of these shall be included in the submittals of this Section.
- C. All exterior piping shall be painted per the Governments requirements.

2.4 ELECTRICAL

- A. Refer to Division 26 Sections for electrical materials, equipment, and installation and testing procedures. Comply with all requirements. Contract drawings identify locations of the power source and available capacity. The contractor is responsible for designing the routing from the source to the equipment connection. Route conductors/feeders on the pipe rack system in conduit.

- B. Each Mobile, Trailer-mounted, Boiler Room shall be provided with lighting and convenience outlets utilizing 120/208V power. Any additional ancillary items that require 120/208V power shall be included in the 120/208V power panel.
- C. All internal wiring shall be sized and installed in accordance with the NEC in EMT conduit. Conduits shall be sized as per NEC and shall permit easy removal of the conductors at any time. All connections to motors and vibrating equipment shall utilize liquid tight flexible metal conduit. Maximum length of the flexible conduit shall be limited to 42" in length.
- D. Light Switches and GFI receptacles shall utilize weatherproof enclosures to prevent any water penetration.
- E. Provide 120 volt convenience receptacles throughout the Mobile, Trailer-mounted, Boiler Room and on interior and exterior walls. Receptacles shall be located on each interior and exterior wall spaced a maximum of 50 feet apart, and within 25 feet of all plant equipment. Additional receptacles served from the emergency system shall be mounted adjacent to every third receptacle served from the normal / standby power system. The emergency wiring system shall be segregated from the normal / standby wiring system in accordance with CEC Article 700.

PART 3 EXECUTION

3.1 DELIVERY

- A. The Contractor shall be responsible for delivery of all equipment to the job site. Equipment shall be FOB jobsite. The Contractor is responsible for obtaining and arranging for any and all permits, escorts, etc. required to deliver the equipment to the job site. Off loading of equipment at the VA's project site is the responsibility of the Contractor.
- B. The Contractor shall provide, to the COR, all off loading information ninety days prior to shipment for coordination purposes. This information shall include but not be limited to the following:
 - 1. Bill of Materials for all components to be off loaded.
 - 2. Weight of all components to be off loaded.
 - 3. Lifting points for all components to be off loaded
 - 4. Any other special rigging instructions required for off loading the equipment.
- C. The Contractor is responsible for equipment delivery and off-loading.
- D. Once off loading has been completed, the Contractor (in the presence of the COR) shall inventory all the equipment and inspect for damage. Upon completion of the inventory the Contractor shall prepare a written letter stating that all equipment included in the contract has been delivered and that no damage has occurred. If any equipment or components are missing or are damaged, it shall be noted in a formal letter to the COR. The letter shall include what components are missing and a schedule for when they will be delivered and/or a plan of action for repairing the damaged components.
- E. The Rental Company shall provide a representative at the site to observe the delivery and off loading of equipment. This representative shall perform the inspection and equipment inventory. This shall be a factory trained representative.

3.2 START-UP SUPPORT

- A. The Contractor shall be responsible for the start up of the equipment. The Contractor shall be present during stack testing of the boiler plant. The Contractor shall provide a start-up plan ninety days prior to commencing start-up.

3.3 TEMPORARY STEAM GENERATING FACILITIES ASSEMBLY

- A. Equipment and control systems shall be shop assembled and wired, and tested to fullest extent possible prior to shipping, so as to minimize field erection requirements.

3.4 TESTING

- A. All non boiler external piping shall be visually and hydrostatically tested in accordance with the ASME B31.1 Power Piping Code. The hydrostatic test pressure shall be as defined in the ASME B31.1 Power Piping Code.
- B. All persons performing visual examinations and evaluating examinations shall be certified according to AWS QC1.

3.5 INSTALLATION, GENERAL

- A. Locate facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.
 - 1. Locate facilities to limit site disturbance as specified in Section 011000 "Summary."
- B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

3.6 TEMPORARY UTILITY INSTALLATION

- A. General: Install temporary service or connect to existing service.
 - 1. Arrange with utility company, COR, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services.
- B. Sewers and Drainage: Provide temporary utilities to remove effluent lawfully.
 - 1. Connect temporary sewers to municipal system as directed by authorities having jurisdiction.
- C. Water Service: Install water service and distribution piping in sizes and pressures adequate for construction.
- D. Water Service: Connect to VA's existing water service facilities. Clean and maintain water service facilities in a condition acceptable to COR. At Substantial Completion, restore these facilities to condition existing before initial use.
- E. Sanitary Facilities: Provide temporary toilets, wash facilities, and drinking water for use of construction personnel. Comply with requirements of authorities having jurisdiction for type, number, location, operation, and maintenance of fixtures and facilities.
- F. Heating and Cooling: Provide temporary heating and cooling required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of low temperatures or high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed.
- G. Isolation of Work Areas in Occupied Facilities: Prevent dust, fumes, and odors from entering occupied areas.
 - 1. Prior to commencing work, isolate the HVAC system in area where work is to be performed according to coordination drawings.
 - a. Disconnect supply and return ductwork in work area from HVAC systems servicing occupied areas.
 - b. Maintain negative air pressure within work area using HEPA-equipped air-filtration units, starting with commencement of temporary partition construction, and continuing until removal of temporary partitions is complete.
 - 2. Maintain dust partitions during the Work. Use vacuum collection attachments on dust-producing equipment. Isolate limited work within occupied areas using portable dust-containment devices.
 - 3. Perform daily construction cleanup and final cleanup using approved, HEPA-filter-equipped vacuum equipment.
- H. Ventilation and Humidity Control: Provide temporary ventilation required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of high humidity. Select equipment that will not have a harmful effect on

completed installations or elements being installed. Coordinate ventilation requirements to produce ambient condition required and minimize energy consumption.

1. Provide dehumidification systems when required to reduce substrate moisture levels to level required to allow installation or application of finishes.
- I. Electric Power Service: Connect to electric power service as shown on the drawings. Maintain equipment in a condition acceptable to the COR.
- J. Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations, observations, inspections, and traffic conditions.
- K. Temporary Fire Protection: Install and maintain temporary fire-protection facilities of types needed to protect against reasonably predictable and controllable fire losses. Comply with NFPA 241; manage fire-prevention program.
 1. Prohibit smoking in construction areas.
 2. Supervise welding operations, combustion-type temporary heating units, and similar sources of fire ignition according to requirements of authorities having jurisdiction.
 3. Develop and supervise an overall fire-prevention and -protection program for personnel at Project site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.
 4. Provide temporary standpipes and hoses for fire protection. Hang hoses with a warning sign stating that hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.

3.7 OPERATION, TERMINATION, AND REMOVAL

- A. Supervision: Enforce strict discipline in use of temporary facilities. To minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.
- B. Maintenance: Maintain facilities in good operating condition until removal.
 1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on a 24-hour basis where required to achieve indicated results and to avoid possibility of damage.
- C. Temporary Facility Changeover: Do not change over from using temporary security and protection facilities to permanent facilities until Substantial Completion.
- D. Termination and Removal: Remove each temporary facility when need for its service has ended, when it has been replaced by authorized use of a permanent facility, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.
 1. Materials and facilities that constitute temporary facilities are property of Contractor.
 2. At Substantial Completion, repair, renovate, and clean permanent facilities used during construction period. Comply with final cleaning requirements specified in Section 017700 "Closeout Procedures."

3.8 TRAINING

- A. The Contractor in coordination with the rental equipment vendor shall provide training for all equipment being provided.
- B. Provide training sessions for twelve (12) VA representative(s) over a period of 7 normal workday(s). Training material shall be provided for each of VA personnel for use during and subsequent to the scheduled training sessions. The training shall be conducted in seven (7) eight (8) hour sessions for the VA's convenience in scheduling personnel attendance. Each of the seven sessions shall cover the information indicated below as a minimum. It is anticipated that each eight hour session will include four (4) hours of classroom training and four hours of in-plant training. Schedule training with COR with at least fourteen days' advance notice. The training shall cover, at a minimum, the following topics:
 1. Train the VA's operation and maintenance personnel on procedures and schedules related to startup and shutdown, troubleshooting, servicing, and preventive maintenance.

2. Review data in the Operation and Maintenance manuals.
3. Review the Sequence of Operations for each plants operation.

END OF SECTION